Stage 3H Physics Note: CCZ4 RHS Block Decomposition

GL-with-AI Project

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1 Scope

Stage 3H is a planning note for the future $4+1$ CCZ4 / SO(3) modified-cartoon right-hand side. It does not implement C++ source terms, add variables, derive the full nonlinear RHS, or prove physical correctness. Its purpose is to split the future RHS into separately checkable blocks before implementation begins.

Unlike the Stage 3C–3G geometry and algebra checks, the full CCZ4 RHS will not have many exact closed-form identities. A passing symbolic check for one piece should therefore be treated as local evidence only, not as validation of the complete evolution.

2 Conventions

The physical spatial dimension is

$$\mathrm{GR\_SPACEDIM} = 4,$$

while the Chombo/grid dimension is

$$\mathrm{CH\_SPACEDIM} = 2.$$

The gridded directions are x, z, and the hidden transverse directions are two equivalent w directions, so

$$N_{\mathrm{hidden}} = \mathrm{GR\_SPACEDIM} - \mathrm{CH\_SPACEDIM} = 4 - 2 = 2.$$

The component h_{ww} is a Cartesian-like hidden conformal metric component, not $g_{\theta\theta}$. The spherical/cartoon reconstruction supplies the external radius factor,

$$g_{\theta\theta} = x^2 \gamma_{ww} = x^2 \frac{h_{ww}}{\chi}.$$

All traces and trace-free projections are full four-dimensional spatial operations. The conformal extrinsic-curvature split therefore uses

$$K_{ij} = \chi^{-1} \left(A_{ij} + \frac{1}{4} h_{ij} K \right).$$

3 Variables

The future geometric RHS blocks are expected to involve

$$\chi, \quad h_{xx}, h_{xz}, h_{zz}, h_{ww}, \quad A_{xx}, A_{xz}, A_{zz}, A_{ww}, \quad K, \quad \Theta, \quad \hat{\Gamma}^x, \hat{\Gamma}^z,$$

plus lapse, shift, and gauge-driver variables as needed. Stage 3H does not choose final C++ enum names or storage order.

The connection-sector variable should use the GRChombo-facing hatted conformal connection,

$$\hat{\Gamma}^i = \tilde{\Gamma}^i + 2Z^i.$$

Thus the future implementation blueprint should evolve $\hat{\Gamma}^A$, $A \in \{x, z\}$, with Z^i understood as the encoded Z4 constraint vector inside $\hat{\Gamma}^i$. Language based directly on Z^i remains useful for constraint-propagation analysis, but the $\hat{\Gamma}^i$ basis avoids an extra translation layer.

4 RHS Blocks

The future implementation should be decomposed into the following blocks:

- $\partial_t h_{ij}$: standard conformal-metric terms, determinant enforcement, hidden h_{ww} evolution, and off-diagonal h_{xz} terms. This block mixes inherited CCZ4 structure with modified-cartoon additions.

- $\partial_t \chi$: trace and gauge terms using the full four-dimensional trace. This block is mostly inherited but depends on the four-dimensional trace bookkeeping.

- $\partial_t K$: trace evolution, lapse Laplacian, Ricci scalar, hidden-sector geometry, and CCZ4 damping terms. Hidden Ricci pieces are modified-cartoon additions and require reference checks.

- $\partial_t A_{ij}$: trace-free Ricci and Hessian blocks, nonlinear A-terms, A_{xz}, and hidden A_{ww}. This is a mixed block with high risk of silently omitted cartoon-only terms.

- $\partial_t \Theta$: Hamiltonian-constraint coupling, constraint-propagation terms, and damping.

- $\partial_t \hat{\Gamma}^A$: hatted conformal connection evolution, encoded Z^i contributions, momentum-constraint coupling, connection terms, hidden-direction divergences, and off-diagonal x, z terms.

- Gauge RHS blocks: lapse, shift, and any driver variables.

- Algebraic enforcement blocks: $\det h_{4D} = 1$, trace-free A, hidden multiplicity, and regularity placeholders.

The gauge block owns the evolution equations for lapse, shift, and any Gamma-driver auxiliary variables. The $\hat{\Gamma}^A$ block owns the occurrences of lapse, shift, and Gamma-driver quantities inside $\partial_t \hat{\Gamma}^A$. This ownership split should prevent double-counting when the blocks are implemented or compared against reference code.

5 Validation Routes

The future checks should be layered:

- Flat or Minkowski data should give a vanishing RHS or known pure-gauge behavior.

- The sheared-flat off-diagonal Ricci gate should protect geometry-source handling, but it is not a full CCZ4 check.

- The diagonal limit $h_{xz} = A_{xz} = \gamma_{xz} = K_{xz} = 0$ should recover the diagonal cartoon formulas.

- Exact unperturbed uniform black-string data should produce a stationary or pure-gauge RHS once gauge conventions are fixed; the gauge-fixed uniform GP-string stationarity and constraint preservation outside turduckening are supporting anchors.
- Hamiltonian, momentum, Θ, and the Z^i vector encoded in $\hat{\Gamma}^i$ should check constraint preservation.
- Symbolic dimensional extraction of the Θ, Z^i, and $\hat{\Gamma}^i$ terms should guard hidden multiplicity and the four-dimensional trace denominator.
- The eventual linearized RHS should reproduce the Gregory–Laflamme threshold/growth spectrum for the SO(3)-symmetric sector after matching radius convention, z-periodic boundary condition, perturbation sector, gauge choice, and extraction variable.
- Separate linearized constraint-violation injection tests around flat or uniform-string data should track Θ, Z^i, Hamiltonian constraint, and momentum constraint response, including κ_1, κ_2, hidden multiplicity, and the 1/4 trace bookkeeping.
- Nonlinear source blocks require later comparison against an external/reference implementation and convergence testing after project conventions are fixed.

The Gregory–Laflamme spectrum is a semi-analytic or literature/numerical spectral anchor from the linear GL eigenvalue problem, not a closed-form elementary identity. Since the physical GL mode is expected to be constraint-satisfying in the continuum, it is a strong check of the physical linearized RHS, K, A_{ij}, metric, gauge, and source-sector couplings. It is not by itself sufficient for the Θ/Z^i damping signs, so a separate constraint-violation damping test is required.

6 Risk Classification

Determinants, inverse metrics, trace-free projections, denominator factors, and diagonal/off-diagonal limits have exact algebraic anchors. Ricci-source terms have useful flat and sheared-flat geometry gates. Full nonlinear CCZ4 damping, constraint propagation, gauge-driver details, enforcement timing, and late-time GL dynamics require external/reference-code comparison and convergence evidence.

7 Non-Goals And Next Stages

Stage 3H does not solve small-x regularity, decide final gauge choices, replace later external/reference comparison, or start implementation. Stage 3I is reserved for small-x regularity and turduckening treatment. Stage 3J is reserved for unit-test fixture design before C++ source terms are added.